Zhan Huanzhe (Joshua)

Atlanta, Georgia | 1(943)206-5809 | hzhan36@gatech.edu | F1 Student

Objective

Dual-degree student in economics (completed) and computer science with demonstrated research experience, including co-authorship of an ICRA paper. Strongly interested in the theoretical foundations of algorithms, computation, control, and robotics, and experienced in translating rigorous analytical thinking into practical implementation. Eager to contribute to research and engineering efforts in robotics, control, and software systems at the earliest opportunity, as preparation for a future PhD path.

Education

Georgia Institute of Technology & Emory University | Atlanta, GA *Fall 2023 – Fall 2027*
Dual Degree Program (Bachelor of Science and Bachelor of Arts)
Georgia Institute of Technology | Atlanta, GA *Fall 2027*
Computer Engineering, Distributed System & Software Design and Robotics & Autonomous Systems
Emory University | Atlanta, GA Fall 2025
Economics Major with Applied Mathematics Minor

Publication

Preprints

P1 Wu, T., Fu, J., Meng, W., Cho, S., **Zhan, H**., & Zhang, F. (2025). Design of a Formation Control System to Assist Human Operators in Flying a Swarm of Robotic Blimps. arXiv preprint arXiv:2505.09511. (accepted by ICRA)

Skills

Programming: Java, Python, C, Rust, Go, Full-stack Web Development, MATLAB
Platforms: Linux (Ubuntu, Arch, RedHat, Debian), Windows (windows, virtual machine, WSL), Other Unix (MacOS)
Hardware: Raspberry Pi, PCB, 3D Printing (FDM, SLA), Laser Cutting, RISC-V boards, Server installation
Software: Latex, OpenCV, Android Studio, Autodesk Series, Adobe Series, GitHub, Blender, ROS
Professional Organizations: National Honor Society, GT IEEE
Communication: Academical essays, design proposals, technical reports, instruction manuals, presentations (large and small audiences), product demonstration
Theoretical: Optimization, Game Theory, Control Theory, Scientific Calculation, Deep Learning, Econometrics, Economy Theory
Languages: Native in Chinese, Fluent in English, Fluent in Hokkien, Intermediate in Japanese

Experience

Hong Kong University of Science and Technology, CKSRI Internship | Hong Kong, China *2024 – Present*
Research Assistant / Robotics Institute
The Cheng Kar-Shun Robotics Institute (CKSRI) is a multidisciplinary platform that supports and facilitates robotics-related research, development and education.

- Co-authored an ICRA 2025 paper by engineering a leader-switching formation control system for robotic blimp swarms.
- Developed a Python/ROS-based distributed control framework, enabling decentralized coordination without external motion capture systems.
- Designed a vision-based relative localization algorithm using monocular cameras, reducing formation tracking error (RMSE) significantly during sharp maneuvers.

Projects

Emory Math Directed Reading Program (DRP)| Emory Math Department *2025*
Student Researcher
Selected for a competitive mentoring program, conducting an in-depth independent study on Information Theory and Theoretical Computer Science under the guidance of a graduate student mentor.

- Conducted an independent study on Information Theory and Shannon Entropy under faculty mentorship through weekly seminars.
- Analyzed the theoretical limits of communication and data compression in distributed computing environments.
- Synthesized complex mathematical proofs and presented a comprehensive technical review on the convergence of information theory and algorithmic complexity to the mathematics department.